



**Mondragon
Unibertsitatea**

**Escuela Politécnica
Superior**

Reinforcement Learning -RL-

Statistical Performance Analysis

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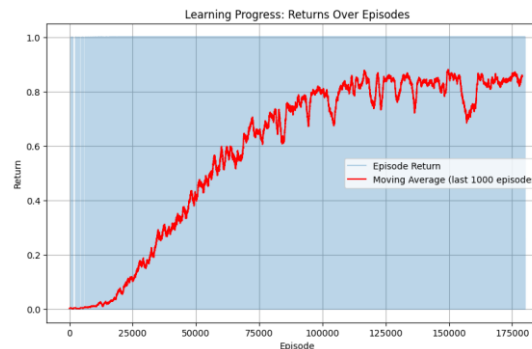
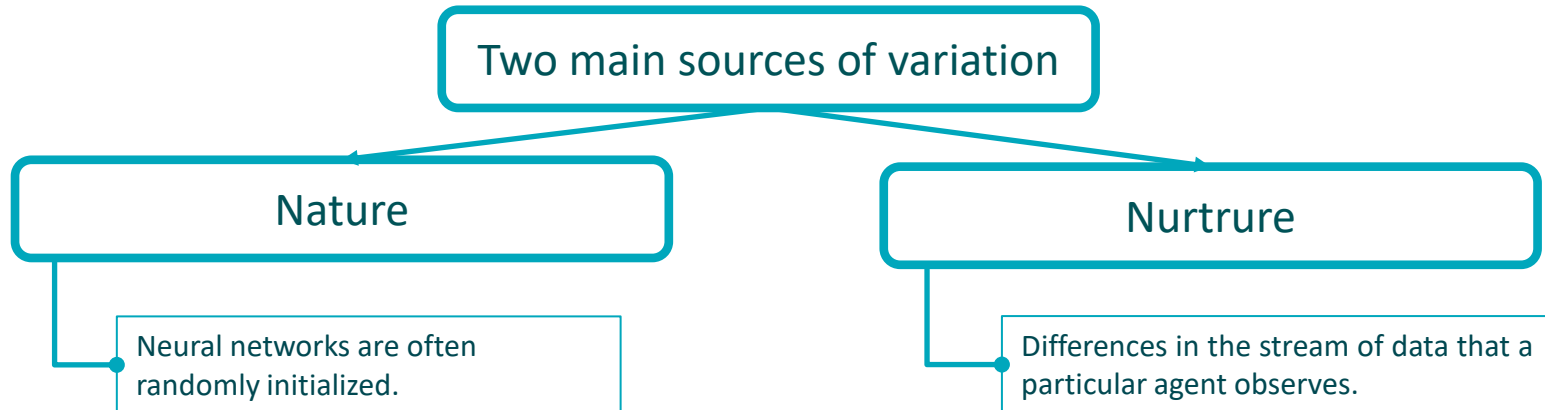
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Agent evaluation

Statistical Performance Analysis

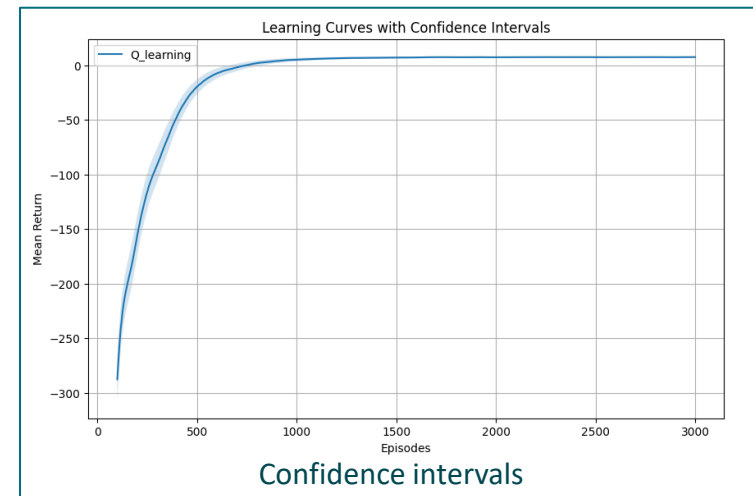
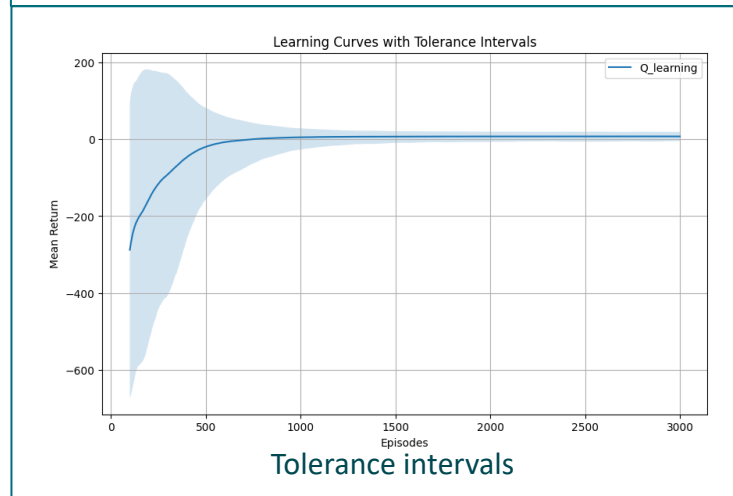
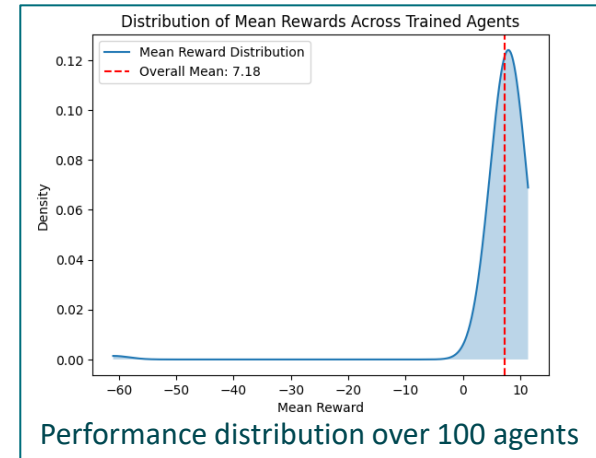
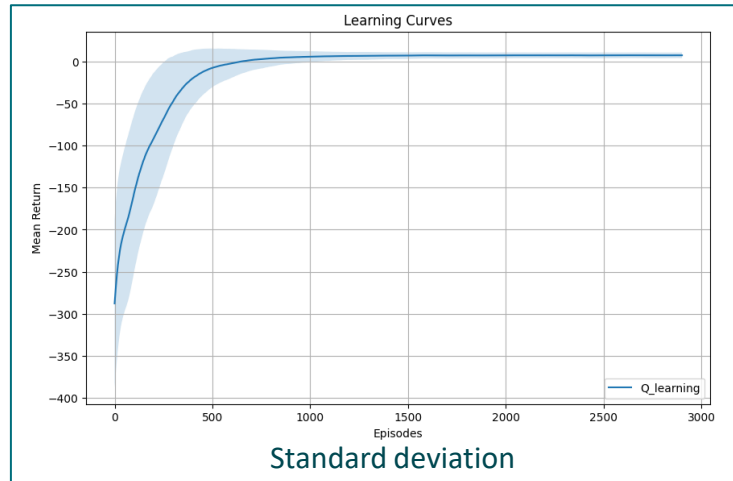
- So far, we have worked with a single execution.
- But it turns out that our algorithms work differently if we run them more than once.



Statistical Performance Analysis

- How we measure the uncertainty?

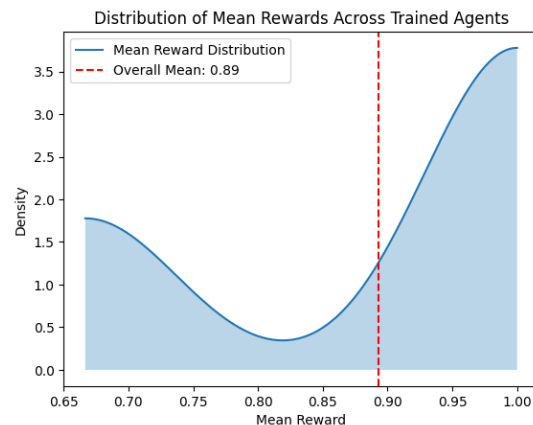
Note: This plots corresponds to Taxi environment using q-learning algorithm



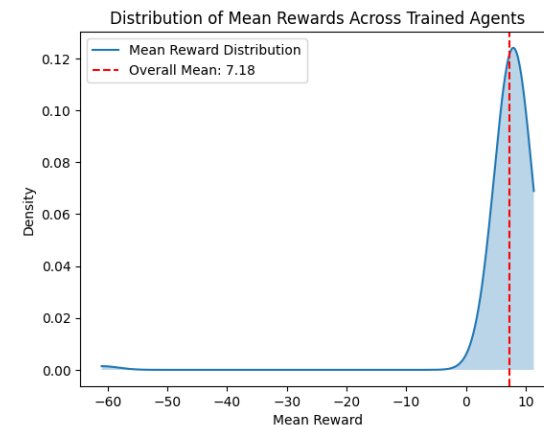
Statistical Performance Analysis

- Evaluation distribution

- In practice, $P(M)$ can be skewed or multi-modal, as we can see in Figure of Frozen Lake. The sample mean (red dashed line) suggests the average agent would achieve a return of approximately 0.9. However, a more detailed explanation would be that any randomly selected agent is likely to receive a return of 1, while some agents will receive a return of 0.65.



Frozen lake 8x8 is_slippery=True
50 runs, 130000 episodes

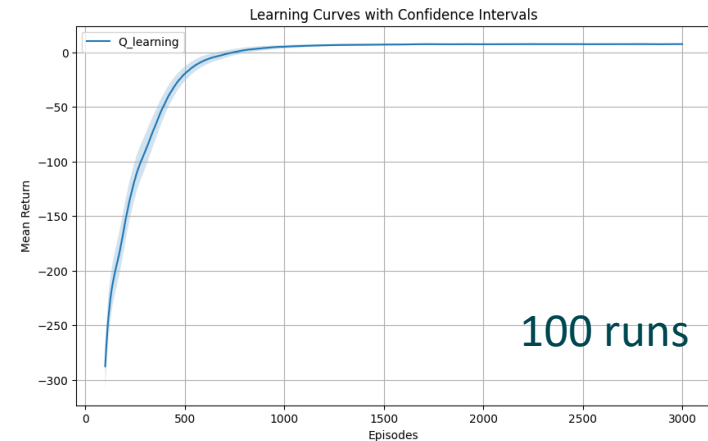
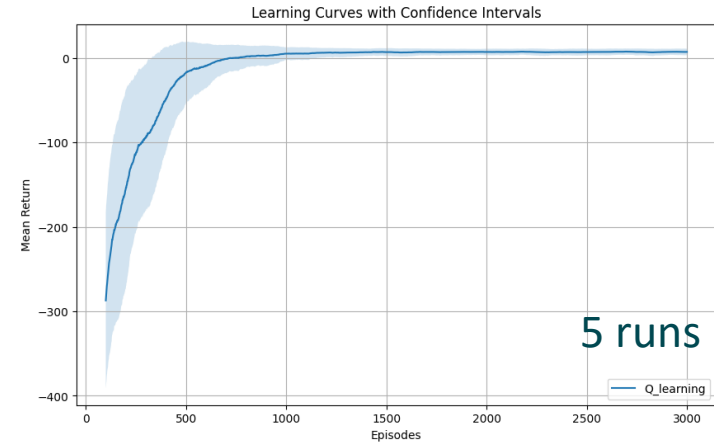
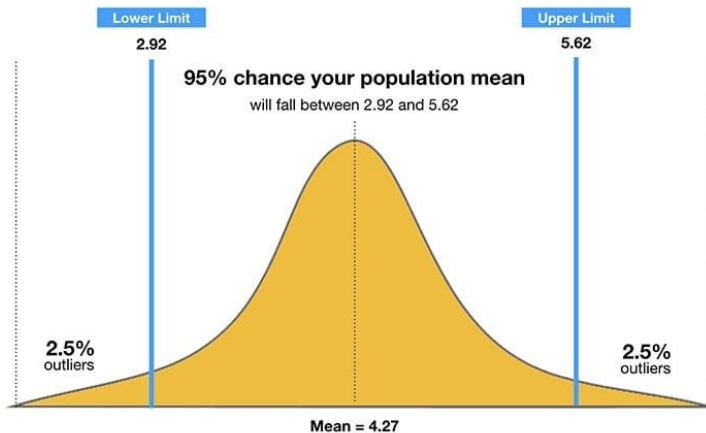


Taxi env
100 runs, 3000 episodes

Note: After the training, each agent was evaluated 3 times, obtaining the mean of the 3 evaluations.

Statistical Performance Analysis

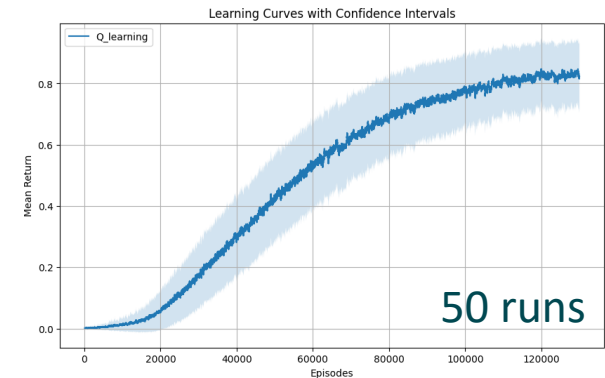
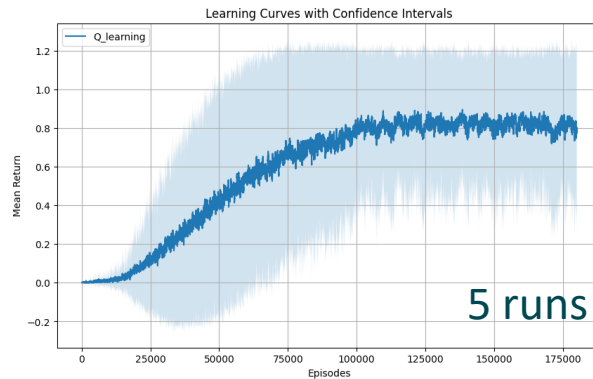
- **Confidence Interval**
 - Show how sure we are about the estimate of a statistic that shows how well an agent is doing. As you collect more and more data, we become more and more confident that our estimate is correct.
 - How confident are we about the mean behavior of an agent?



The more runs, the more confident we are of the behavior

Statistical Performance Analysis

- Example
 - Confidence interval in Frozen Lake 8x8 Stochastic Environment

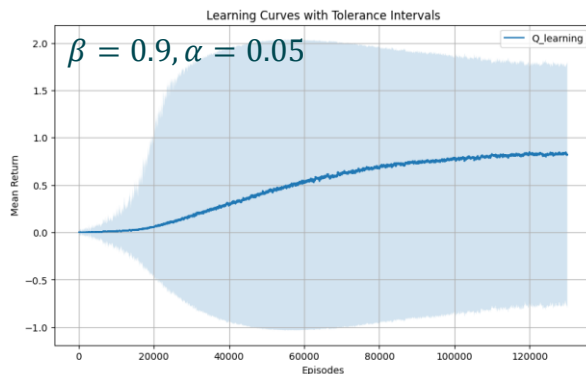


50 runs with 130_000 episodes each, 6_500_000 episodes!

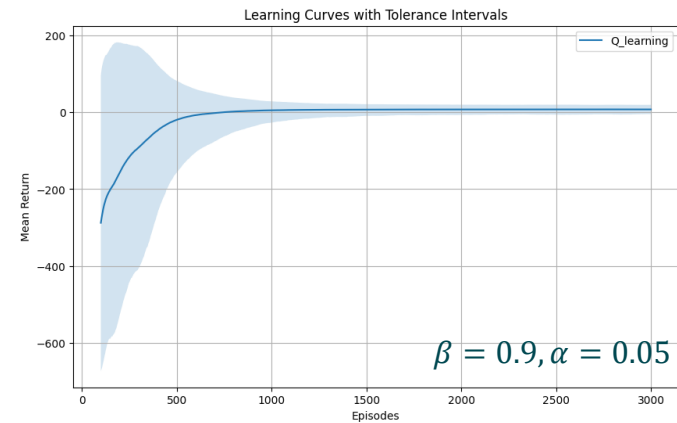
Statistical Performance Analysis

- Tolerance Interval

- Tolerance intervals attempt to capture the variation in the distribution of performance.
- A tolerance interval is about where most of your data (runs/episodes) lie, rather than just how confident you are in the average return.



Frozen lake 8x8 is_slippery=True
50 runs, 130000 episodes



Taxi env
100 runs, 3000 episodes

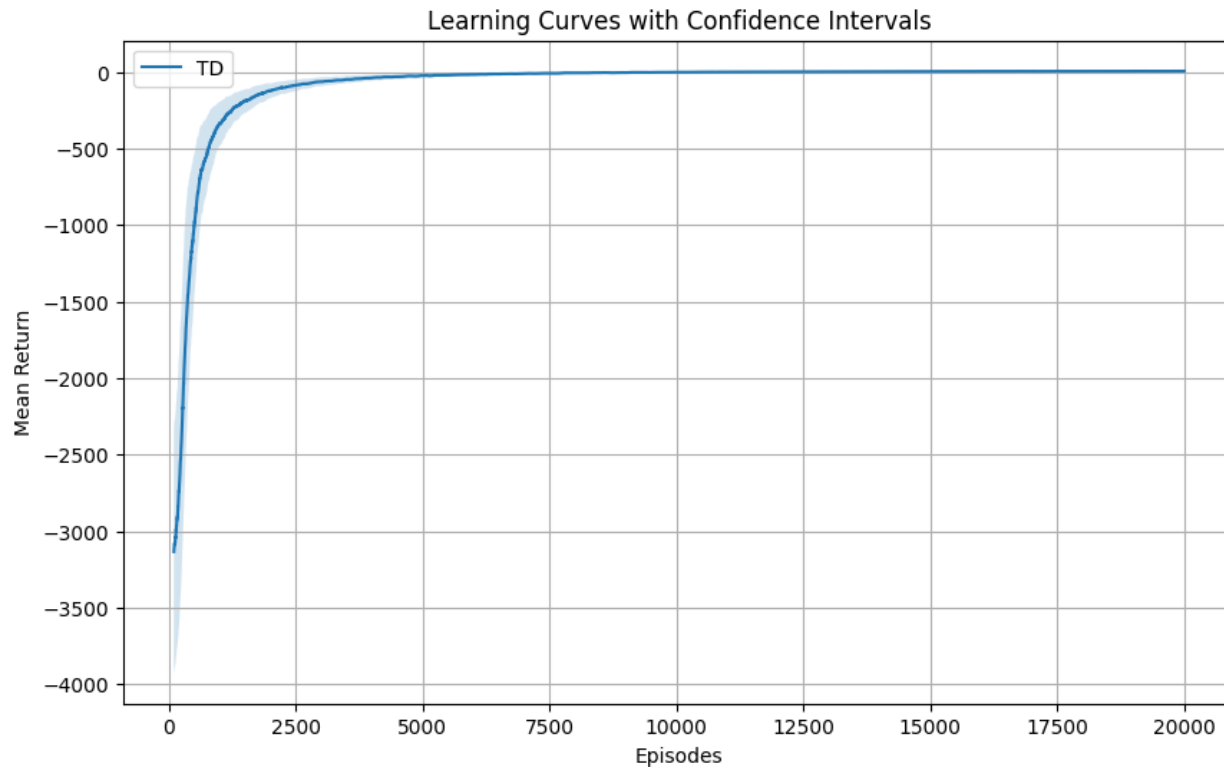
So, with a 95% of confidence I am sure that 90% percent of experiments will be between these boundaries

Statistical Performance Analysis

- Wait... but what of them should I use?
 - You might report a mean (average) return with a confidence interval and a standard deviation to show both your estimate of the mean and the spread.
 - If you need to ensure most runs (not just the average) are good, you might also include a tolerance interval.

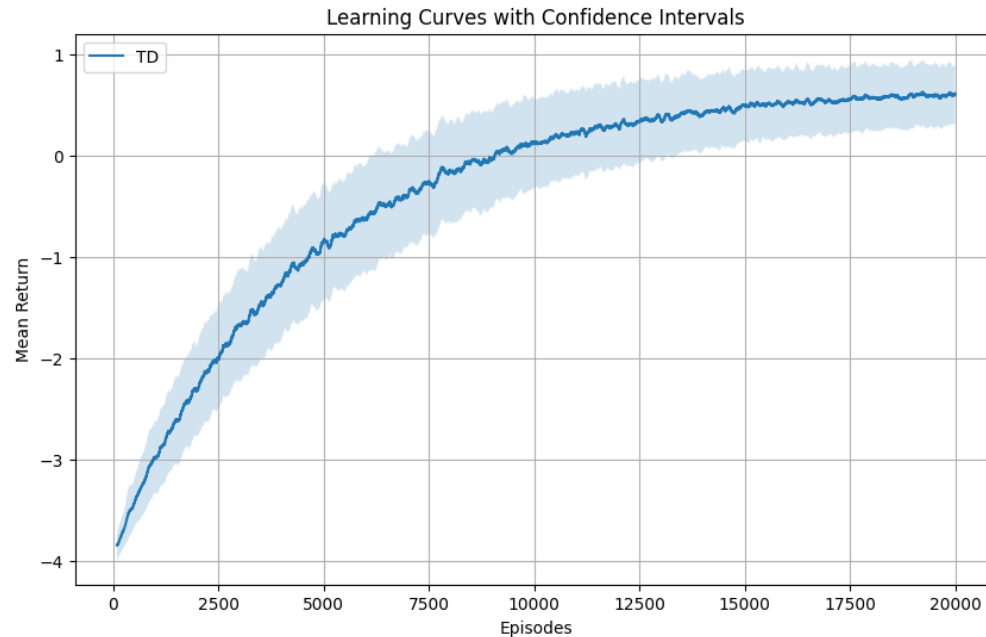
Statistical Performance Analysis

- So far, we have used return over episodes to measure learning efficiency. But is this the best metric?



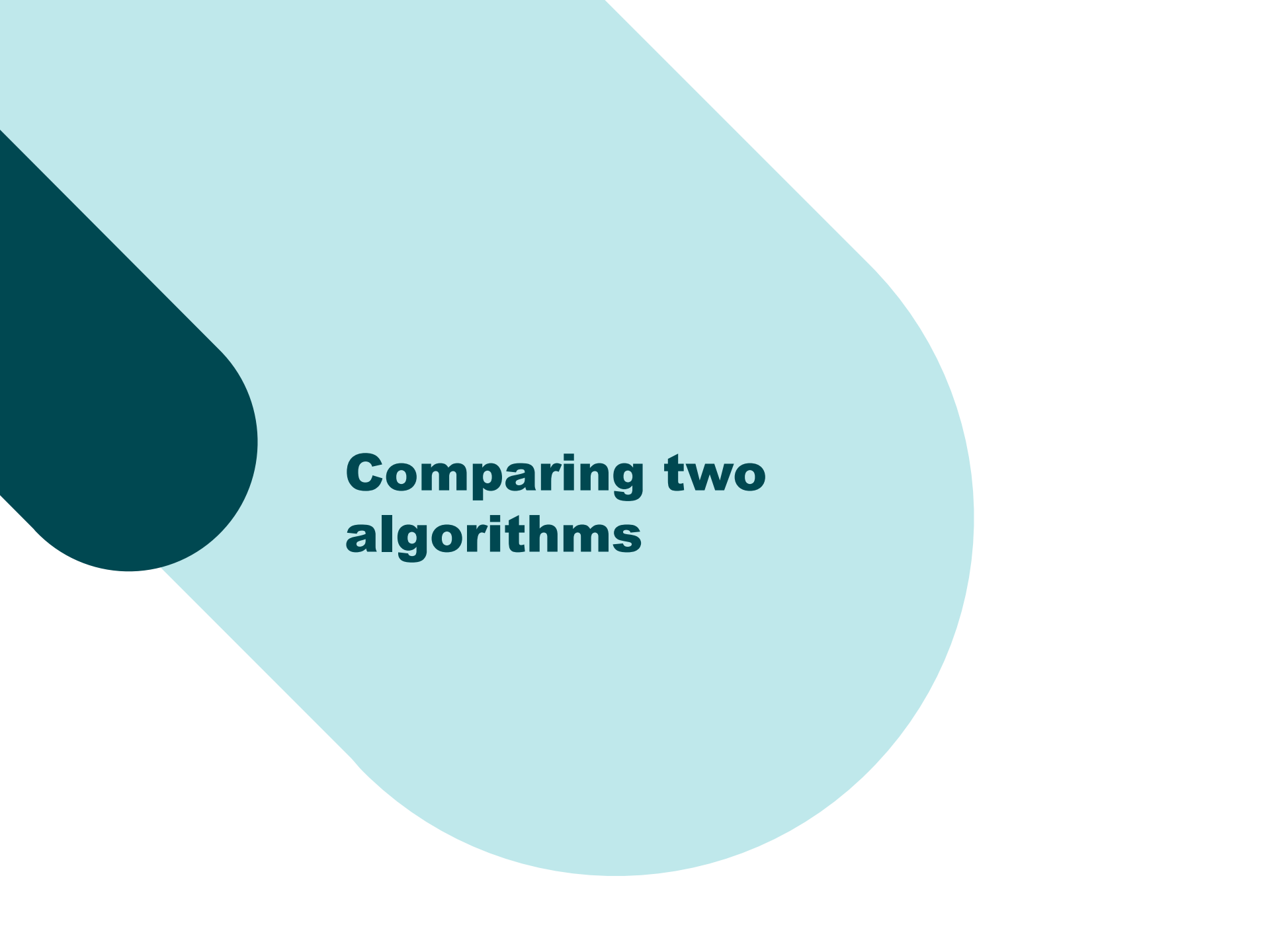
Statistical Performance Analysis

- What about return per step over episodes?



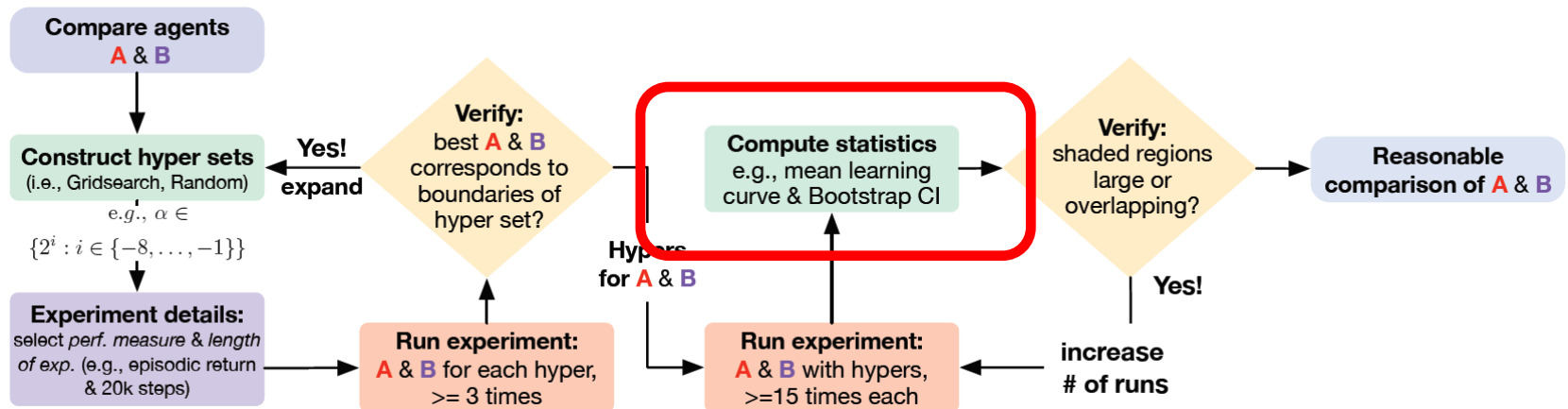
Statistical Performance Analysis

Metric	Tracks	Best for	Pros	Cons
Return per episode	Total reward per episode	Overall learning progress	Simple, intuitive	Doesnt consider efficiency per step
Return per step over episodes	Return normalized by steps per episode	Learning efficiency evaluation	Allow fair comparison accross episode lenghts	Can be misleading if episode lenghts vary too much
Return over steps	Cumulative return vs. Total steps	Fine grained learning analysis	Tracks progress independent of episode structure	Noisy, harder to interpret without smoothing



Comparing two algorithms

Comparing two algorithms



Patterson, A., Neumann, S., White, M., & White, A. (2023). Empirical Design in Reinforcement Learning. *Journal of Machine Learning Research*, 25, 1–63.

Comparing two algorithms

- **Welch t-test**

- Helps analyzing difference on means with the assumption that each of the groups have different variances.
- RL algorithms have stochastic behavior: results vary across runs.
- We often compare performance metrics like:
 - » Return per episode (total reward per training episode).
 - » Return per step (efficiency of learning).

$$T = \frac{\overline{X_1} - \overline{X_2}}{\sqrt{\frac{S_1^2}{N_1} + \frac{S_2^2}{N_2}}}$$

- Interpreting T-Statistic & P-Value

- **Large $|T|$** → Greater difference between algorithms.
- **P-Value < 0.05** → Strong evidence that one algorithm significantly outperforms the other.
- **If P-Value > 0.05** → Differences may be due to randomness, not algorithm superiority.

Comparing two algorithms

- Taxi env. (Welch t-test)
 - This is the same experiment carried out on the taxi environment.
10 runs

Return per episode

Welch's T-Test Results:

T-Statistic: 11.061 **P-Value:** 1.469e-06

Temporal Difference - Mean: -80.918, **Std:** 0.869

Monte Carlo - Mean: -140.110185, **Std:** 16.899

Return/steps per episode

Welch's T-Test Results:

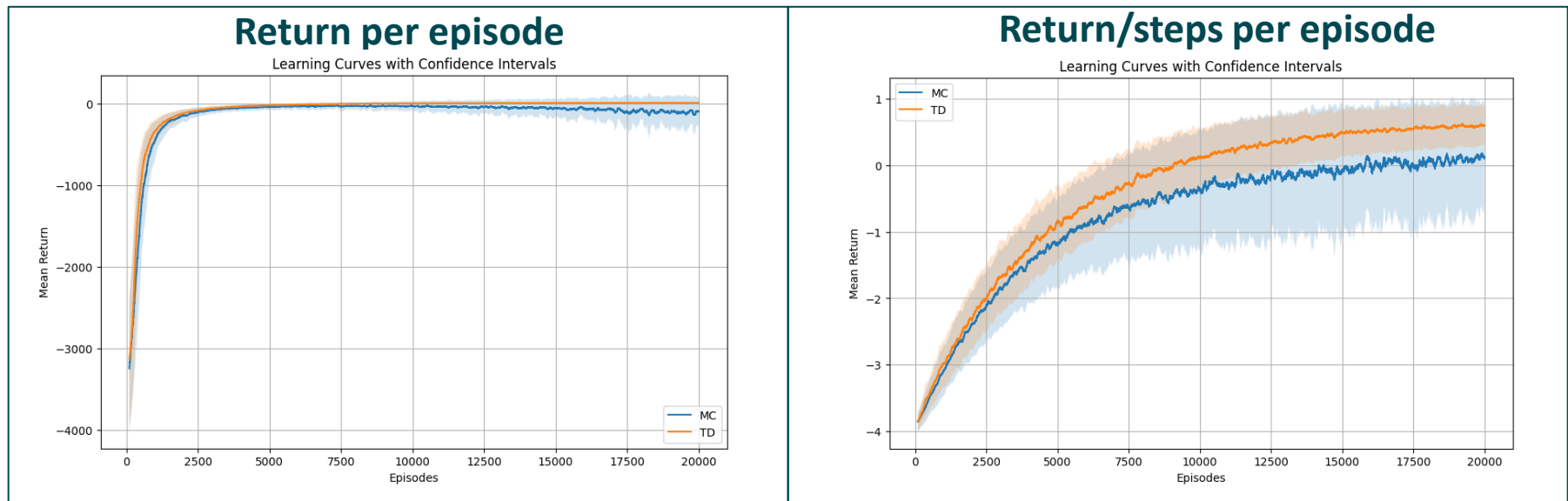
T-Statistic: 9.814 **P-Value:** 4.121e-06

Temporal Difference - Mean: -0.376, **Std:** 0.0038

Monte Carlo - Mean: -0.760, **Std:** 0.123

Comparing two algorithms

- Taxi env. (Confidence interval)
 - This is the same experiment, carried out on the taxi environment.
10 runs

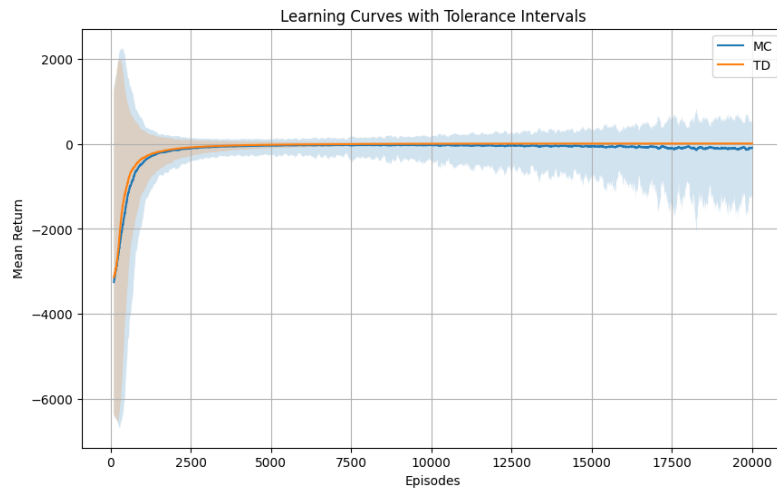


Which one should we use?

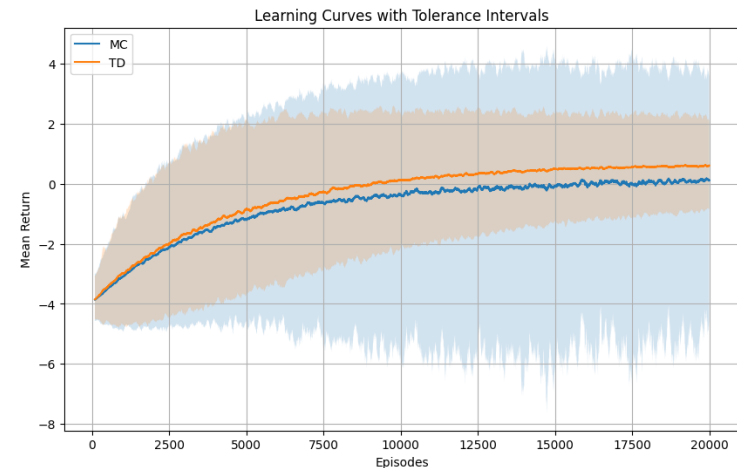
Comparing two algorithms

- Taxi env. (Tolerance interval)
 - This is the same experiment carried out on the taxi environment.
10 runs

Return per episode



Return/steps per episode



Eskerrik asko
Muchas gracias
Thank you

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